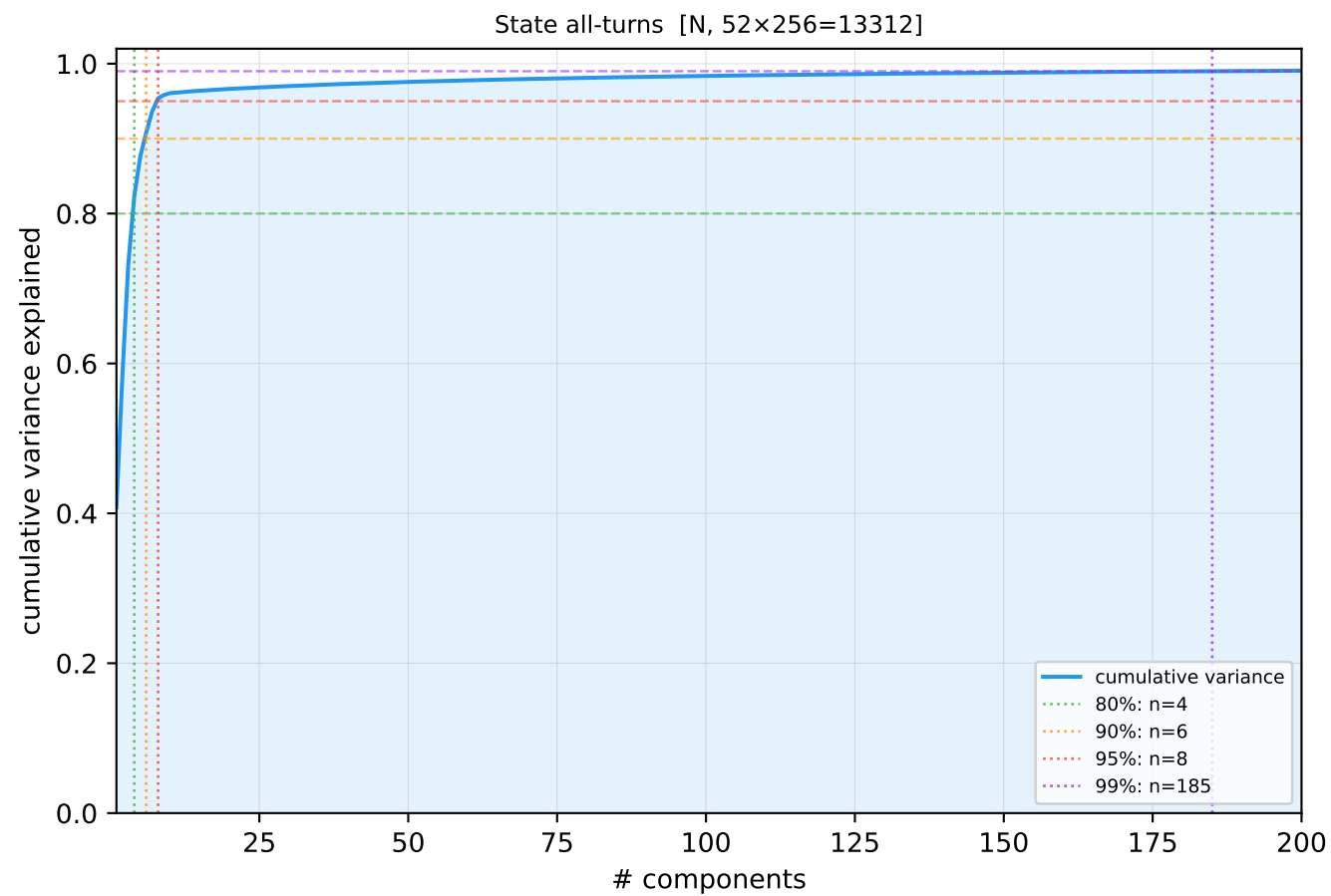
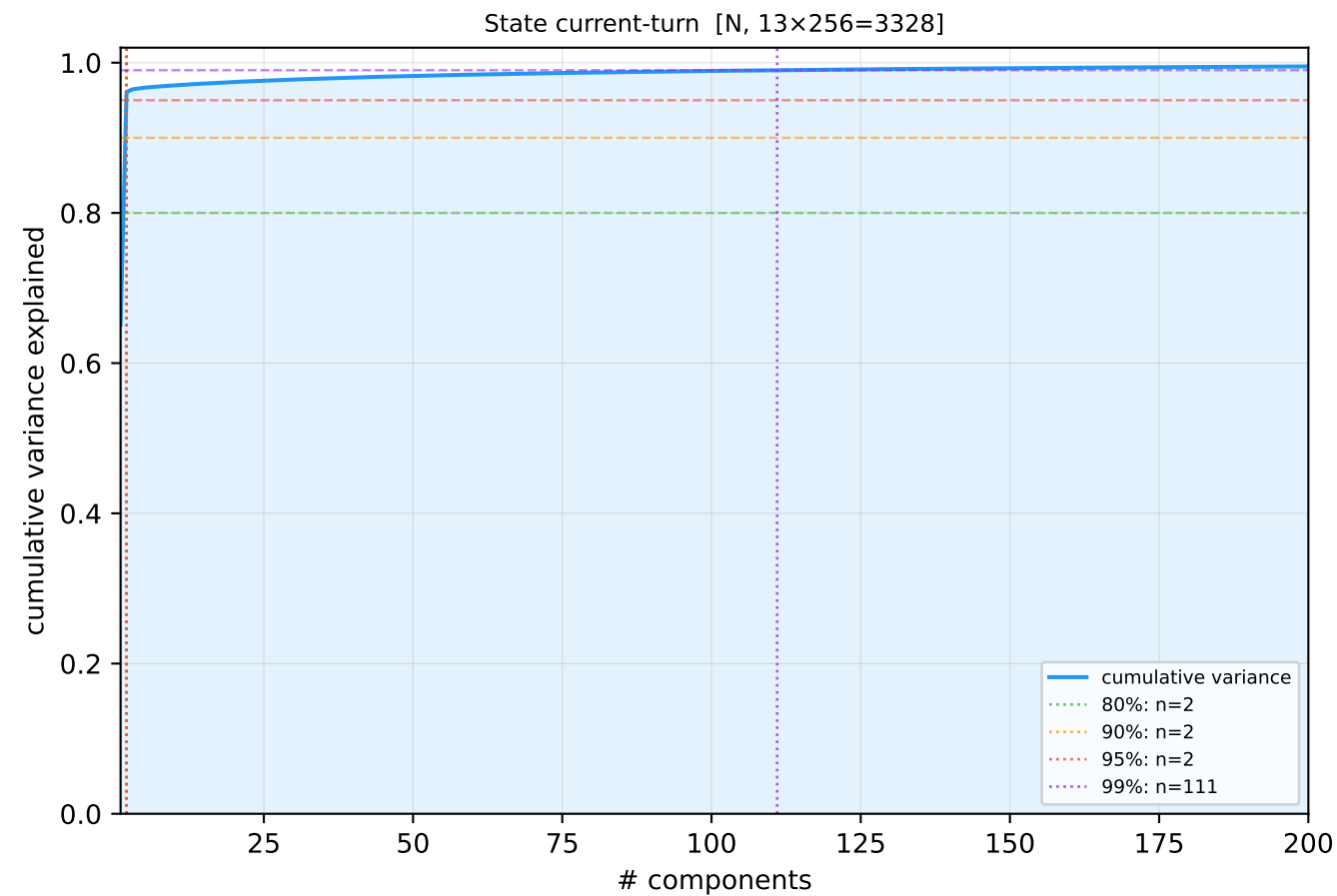
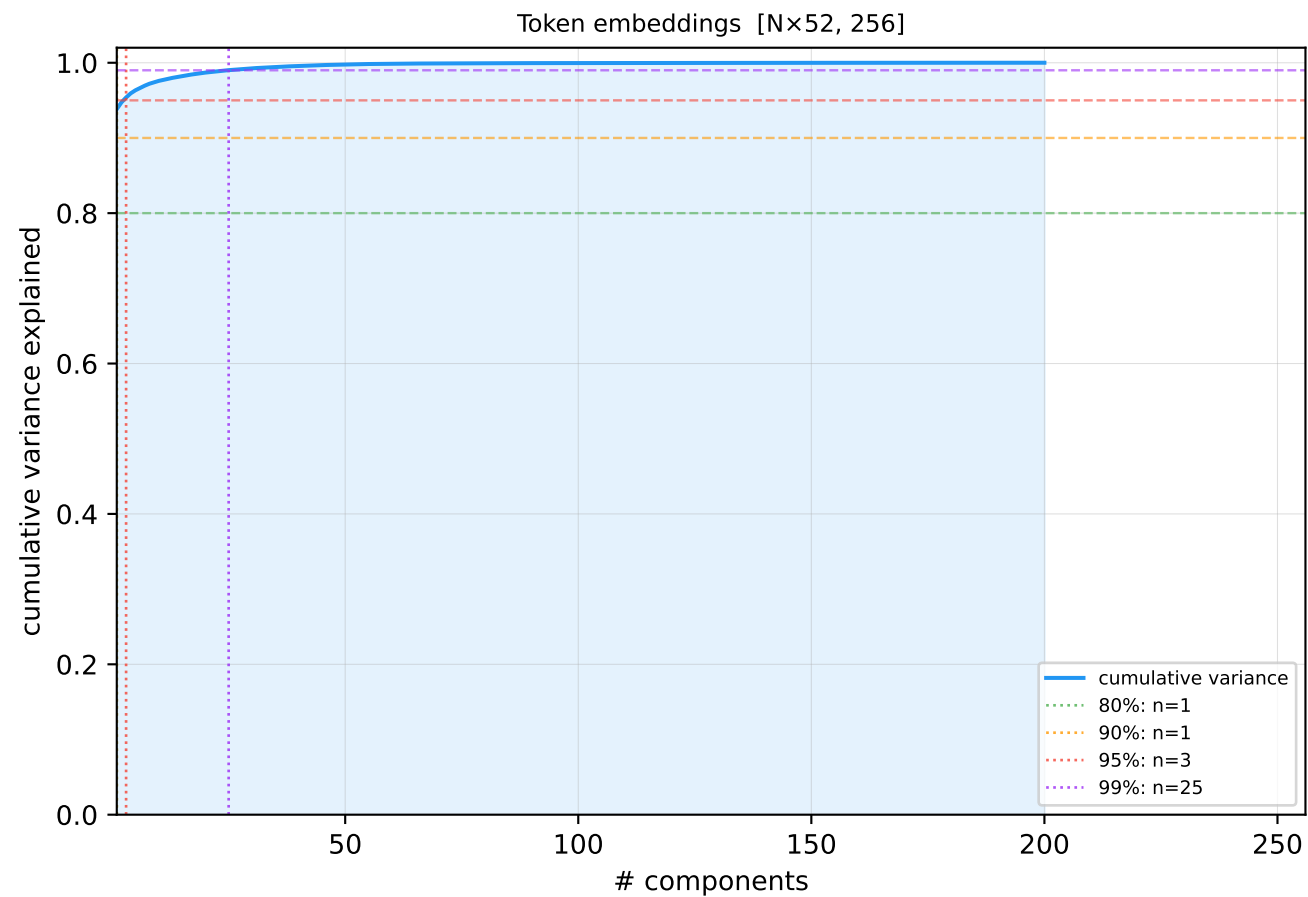


CynthAI_v2 — PCA / TruncatedSVD: effective dimensionality



Summary: components needed

Level	80%	90%	95%	99%
Token emb (256D)	1	1	3	25
State cur (3328D)	2	2	2	111
State all (13312D)	4	6	8	185

n_components to reach threshold
 (TruncatedSVD, mean-subtracted data)

token_emb: each of the N×52 tokens
 state_cur: concat of 13 current tokens
 state_all: concat of 52 all-turn tokens